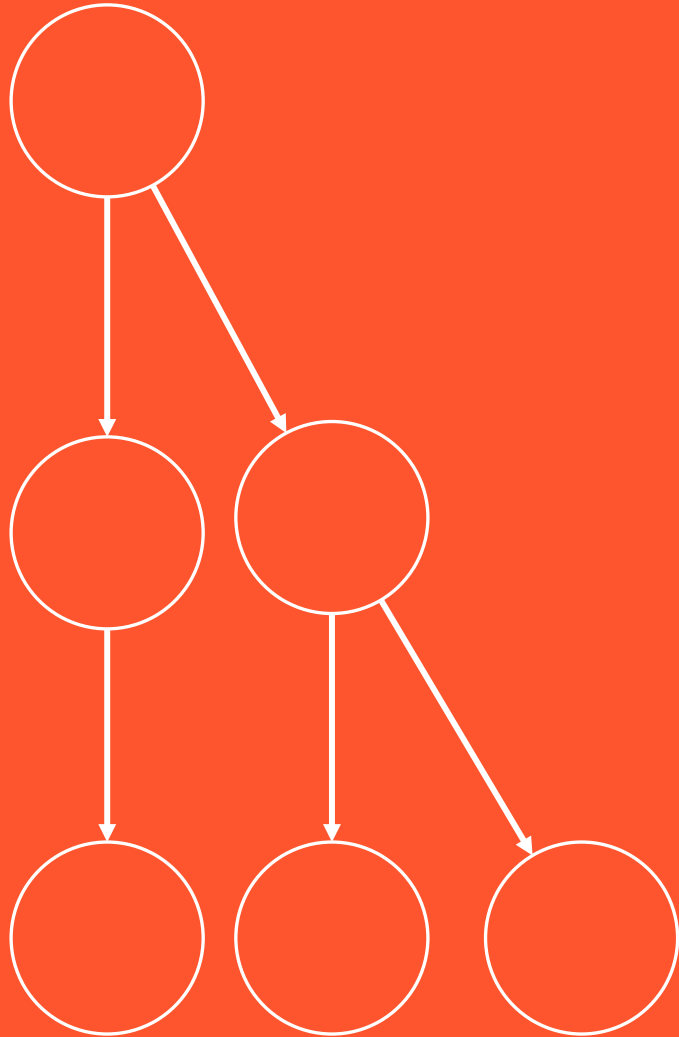




BCOG 100

Comparative Approaches

Lecture 2.2 — Comparisons Across Species



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Why This Is Even Allowed



Darwin, *The Descent of Man* (1871): the difference between the human mind and the minds of other animals is one **of degree**, not **of kind**.

That claim is what makes the whole enterprise legitimate. If minds are products of evolution, then minds have histories — and related species should have related minds.

It is a substantive claim that somebody argued for. **It is not unanimous.**

Natural Selection, From Scratch

Three ordinary observations. One guaranteed outcome.

Variation

individuals are not identical

+

Inheritance

offspring resemble their parents

+

Differential reproduction

some variants leave more
surviving offspring



The variants that reproduce more become more common.

Not likely — guaranteed.

and then it happens again, every generation

Not a Force. Bookkeeping.



Before industrial Britain: the peppered moth was pale and speckled, nearly invisible on lichen-crusted bark.

Coal smoke killed the lichens and blackened the bark. Within decades, dark moths dominated the sootiest regions. Clean-air laws reversed it again.

The soot did not darken the moths. It changed which moths a bird could see.

Same shape, present tense: the antibiotic did not *make* the bacteria resistant. It killed everything that was not.

Selection edits an existing range of variation. It does not create it.

No Foresight, No Progress

Selection cannot plan. It cannot aim at a future benefit, and it cannot hold out for a better design that would require passing through a worse one first.

So an evolutionary **why** is never a claim about what something is *for*, the way a hammer is for driving nails. It is a claim about **the past** — about what happened to work for a creature's ancestors.

An **adaptation** is a trait that became common because the organisms that had it were more likely to survive and reproduce *in the environment they were actually living in*.

Local fit. Not improvement. When circumstances change, the fit can vanish.

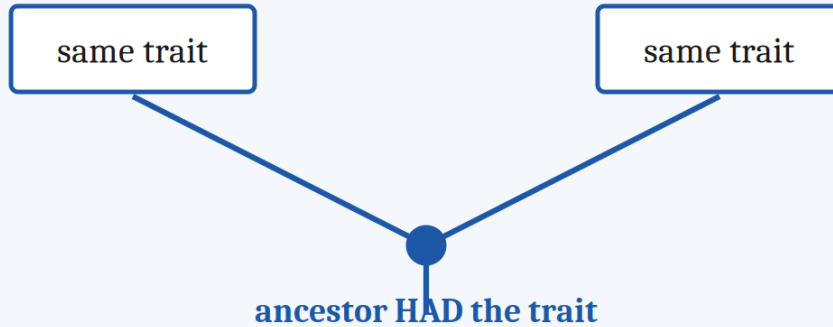


Two Names for Two Histories

The tips are identical. The histories are not.

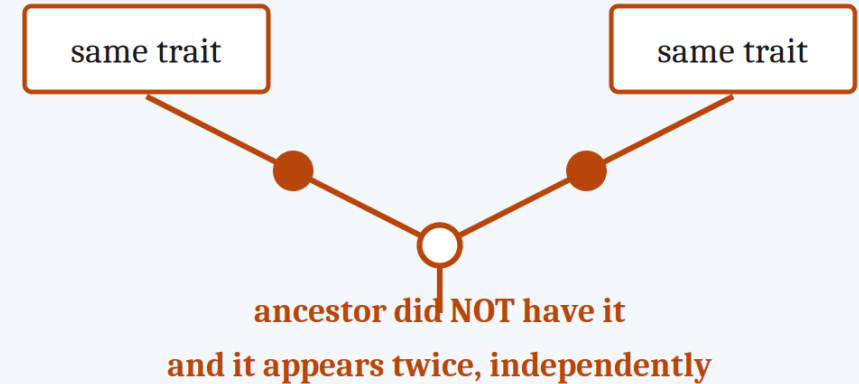
Homology

inherited from a common ancestor



Analogy

arose separately, under similar pressure

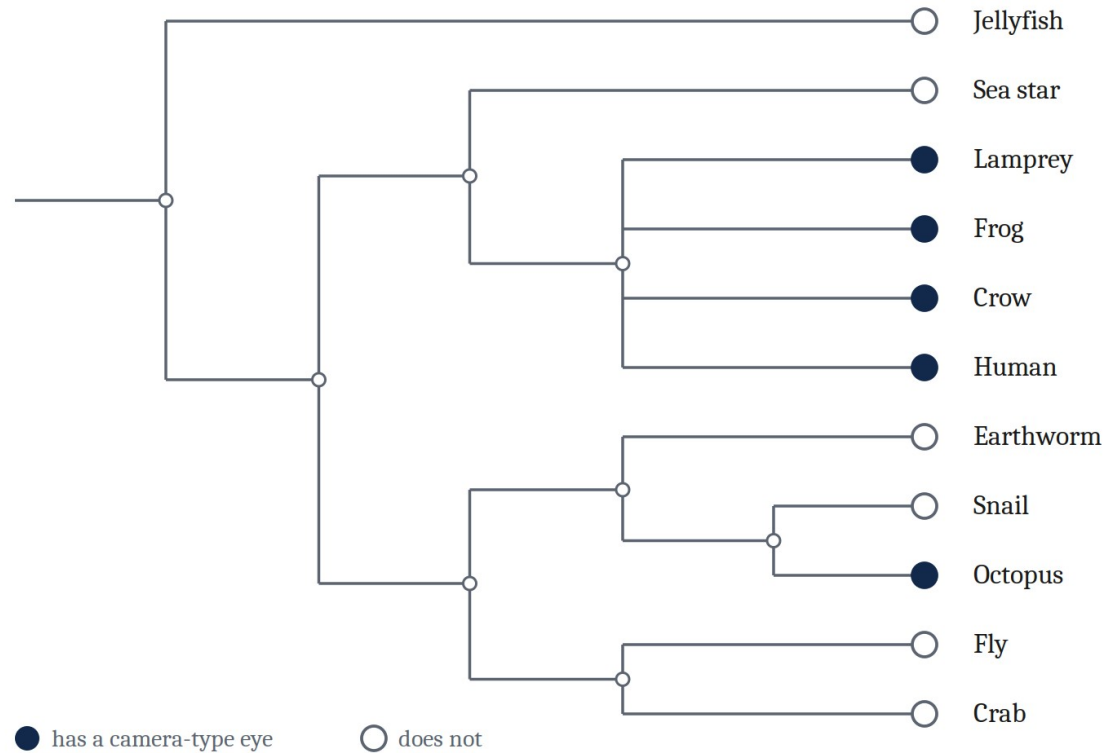


Nothing about how a resemblance LOOKS will tell us which of these it is.

Convergent evolution is the process that produces the right-hand case.

A **common ancestor** is a species from which both are descended.

The Tree Is a Procedure



A **phylogeny** is a species' evolutionary history. A **phylogenetic tree** is the map of it — relatedness by descent, with branch points marking common ancestors.

Close together, both have it → inheritance is the natural first hypothesis.

Far apart, and the species in between lack it → inheritance is ruled out. Convergence or coincidence.

A **selective pressure** is a feature of the environment that makes one variant reproduce more successfully, so that variant spreads.

Human and octopus camera eyes are **analogous**. The genetic switch that starts an eye developing is **homologous** — inherited from an ancestor with no eyes at all.

It Happened More Than Once

New Caledonian crows manufacture tools — trimming twigs into hooks to fish grubs out of wood. In the wild, untaught.

Last common ancestor of birds and mammals: a reptile-like vertebrate, roughly **300 million years ago**, before the dinosaurs, with nothing about it suggesting tool manufacture.

Run the procedure: corvid and human tool use is **analogous**.

Scrub jays remember not only *where* they cached, but *what*, and *how long ago* — recovering the perishable items first. Memory for a specific past event, long assumed to be ours alone.



It Happened More Than Once

The **octopus** is the more radical case. A large nervous system arranged nothing like ours — most of its neurons in its arms rather than centralized. Our lineages separated roughly **600 million years ago**, before anything we would call a brain.

Godfrey-Smith: meeting an octopus may be the closest we will come to meeting an intelligent alien.



Why Convergence Is Evidence

The worry. Module 1 said the computational level asks what problem a system solves — and for living things the answer is adaptive. But adaptive stories are cheap.

Humans detect snakes fast → an anti-venom adaptation. Obviously.

Suppose we detected them slowly → freezing is better once a predator is close, and rapid searching is the movement that draws a strike. Also obviously.

An explanation that would have accommodated either outcome has not told us anything.

The answer. When two lineages sharing no history arrive at the same solution, the problem they were both solving was **real**. The same answer reached twice is a strong hint the question was not one we made up.

The fence. "The same capacity" is a claim to be tested, not read off a resemblance. Test it by finding out **whether they break the same way**.

There Is No Ladder

The picture in everyone's head

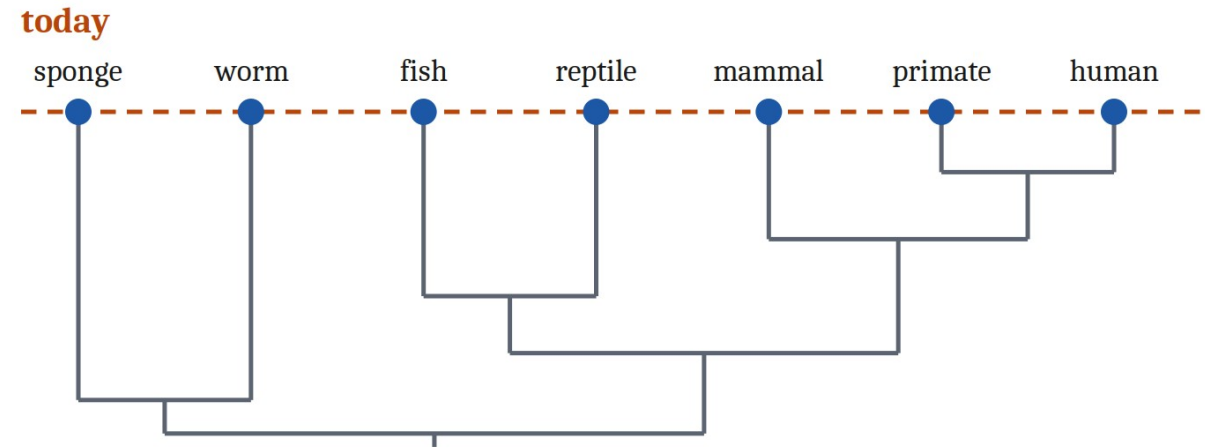
inherited from the medieval great chain of being



ranked by nearness to us

What is actually the case

every living species, the same span of time



Every tip ends on the same line. Nothing is further along.

schematic — branch order is illustrative, not a published phylogeny

So “is a tick smarter than a crow” is not a hard question. It is a badly formed one.

Judge a capacity against the world the animal actually lives in — von Uexküll's **Umwelt**, the environment as that species senses and acts on it.

A tick's is built out of butyric acid, warmth, and touch. That is the whole world, and within it the tick is superb.

Both Errors, One Cause

Clever Hans, Berlin, early 1900s. Asked "what is three plus five," the horse tapped a hoof the correct number of times. Reliably. Taken as evidence that horses understood language and arithmetic.

Pfungst worked out what was happening: Hans was reading tiny, unintentional changes in the posture and breathing of whoever in the room knew the answer.

That is **anthropomorphism**. Its opposite is de Waal's **anthropodenial** — refusing an animal a capacity it plainly has, because granting it feels unscientific.

Both errors have the same cause: deciding in advance what the answer should be.

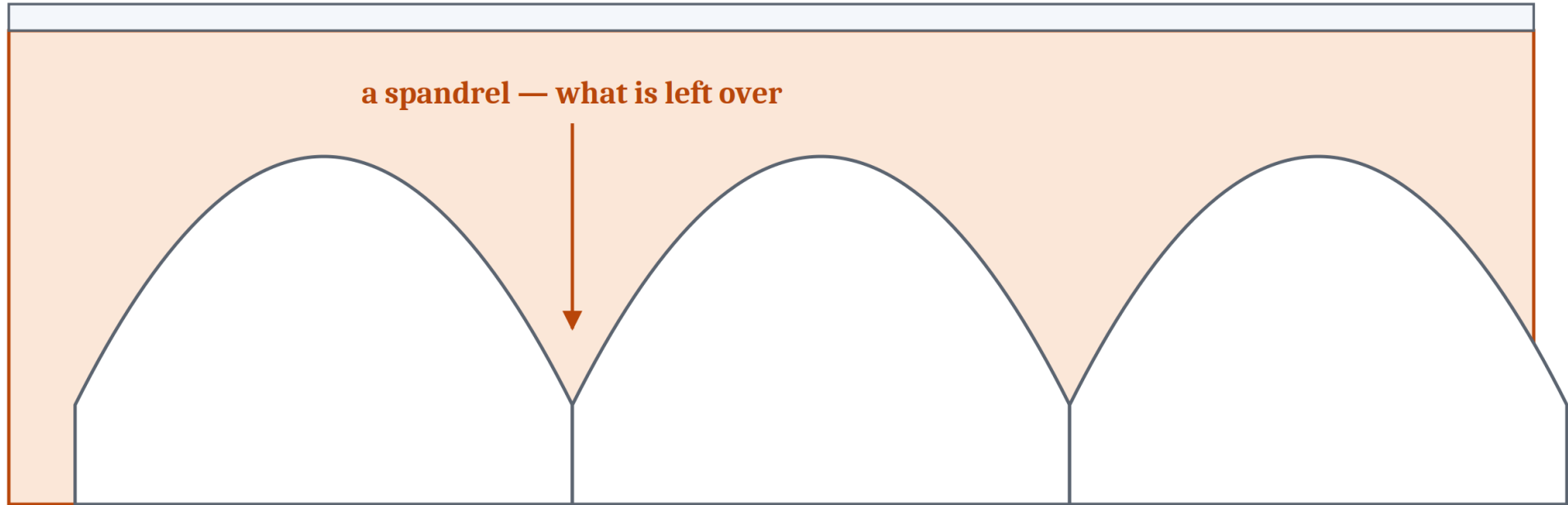
Morgan's Canon (1894): do not explain a behavior by a higher mental process if a simpler one accounts for it. This is the founding discipline of **ethology** — the study of animal behavior as a biological trait with an evolutionary history.



Stories Written to Fit

Put a dome on arches, and you get these spaces. Nobody chose them.

the dome sits on this



Later builders found them excellent places for paintings.

Study only the paintings, and you conclude the space exists FOR the paintings — exactly backwards.

Kipling wrote how the leopard got his spots and the camel his hump. Each story fits its animal perfectly — **which is exactly the problem. Each was written to fit.**

The corrective — why is not one question. Tinbergen (1963) counted four — what causes it now • how it develops over a lifetime • what it is good for • how it evolved.

The Spine — and the Objection to It

These arguments are never neutral. Whether an animal thinks, plans, or suffers bears on how we may treat it — as a laboratory subject, as a farmed animal, as a pet. Arguments about animal minds carry a charge that arguments about animal digestion do not.

From Module 3 onward, this is the structure of the course. Each module takes up a step in the evolution of the human mind and brain, starting before neurons existed. We understand what a capacity is *for* by understanding the problem it solved — and the way to see the problem is to look at who else solved it.

The obvious objection: isn't that the ladder you just rejected?

No. Evolution has no destination and no lineage is further along than any other. We follow the human lineage because human cognition is what this course is trying to explain — **not because it is best, and not because it is last.**

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